

AI Idea Forge

AI Product Discovery Platform

– Batch 4 Group 8 –

CHAPTER

Week 3 – System Integration & Workflow Automation

Document Title:	Week 3 – System Integration & Workflow Automation
Prepared By:	Batch 4 – Group 8
Audience:	University Faculty – Technical Implementation Review
Date:	August 1, 2026

Table of Contents

1. Introduction	4
2. Overall System Architecture	4
3. End-to-End Workflow	6
4. Initial Analysis Workflow (SS1)	7
5. AI Agent Workflow (SS2)	8
6. Generic AI Agent Workflow	9
7. AI Agent Responsibilities	10
8. Individual AI Agent Documentation	11
8.1 Market Research	11
8.2 Competitor Analysis	11
8.3 Gap Analysis	12
8.4 Feasibility Analysis	12
8.5 Risk & Challenge	13
8.6 Recommendation Engine	14
9. Lovable ↔ n8n Integration	15
10. n8n ↔ Supabase Integration	15
11. Workflow Automation	17
12. Error Handling	17
13. Design Principles	18
14. Conclusion	18

Appendix A – AI Prompt Engineering	19
A.1 Market Research Prompt	19
A.2 Competitor Analysis Prompt	20
A.3 Gap Analysis Prompt	21
A.4 Feasibility Analysis Prompt	22
A.5 Risk & Challenge Prompt	23
A.6 Recommendation Engine Prompt	24
Appendix B – JSON Output Schemas	26

1. Introduction

This chapter documents the system integration and workflow automation architecture that powers the AI Idea Forge platform. Where earlier chapters addressed interface design and database structure, this chapter focuses on how those layers are connected: how a startup idea submitted by a user is carried, transformed, and enriched as it moves through the platform's automated AI pipeline.

Workflow automation is required because the platform's core value — a multi-domain AI analysis of a startup idea — depends on coordinating several independent, asynchronous processes: validating user input, invoking a large language model, parsing its response into a strict structured format, persisting that result to the correct database table, and keeping the front-end interface in sync with the analysis as it progresses. Performing this coordination manually, or embedding it directly in application code, would be difficult to maintain, difficult to extend, and fragile in the face of AI response variability. A dedicated automation layer keeps this orchestration logic visible, modular, and independently testable.

Five technologies form the backbone of this architecture:

- **Lovable** — the front-end application layer through which users submit ideas and view analysis results.
- **n8n** — the workflow automation engine that orchestrates every AI call, data transformation, and database operation in the pipeline.
- **Supabase** — the PostgreSQL-backed data layer that stores submissions, AI-generated master records, and every specialist agent's output.
- **AI Model** — the large language model (Llama 3.1 8B Instant, served via Groq) that performs the actual domain analysis for each stage.
- **REST APIs** — the HTTP contract (webhooks and structured JSON payloads) that connects Lovable, n8n, the AI model, and Supabase into a single coherent pipeline.

2. Overall System Architecture

The diagram below presents the platform's overall system architecture as a layered pipeline, from user input through to the final presentation of results. Each stage hands off a well-defined artifact to the next, which keeps the responsibilities of each layer isolated.

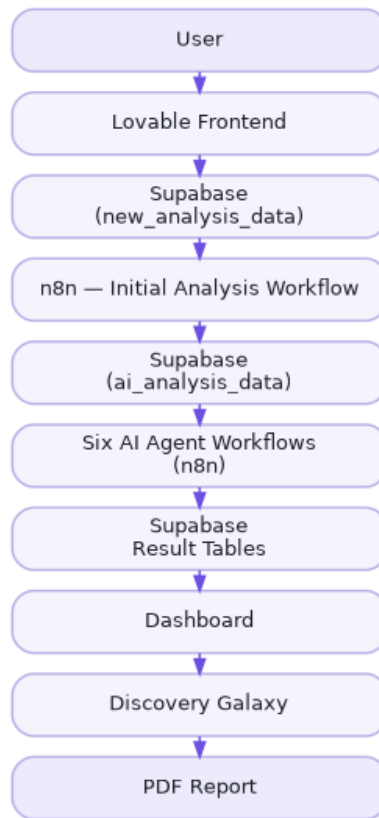


Figure 1. Overall System Architecture – AI Idea Forge Pipeline

Component	Role
User	Initiates the process by submitting a startup idea through the Lovable front end.
Lovable Frontend	Provides the submission form, Dashboard, and Discovery Galaxy interface; performs input validation before forwarding the request.
Supabase (new_analysis_data)	Persists the raw, validated idea submission as the entry point of the data model.
n8n — Initial Analysis Workflow	Retrieves the raw submission, calls the AI model, and produces the consolidated master record.
Supabase (ai_analysis_data)	Stores the AI-generated title, description, industry classification, and matching score for the idea.
Six AI Agent Workflows (n8n)	Independently analyze the idea across six business domains using a shared, reusable workflow template.
Supabase Result Tables	Stores each agent's structured findings in its own dedicated table.
Dashboard	Reads consolidated scores and status across all agents to give the user an at-a-glance summary.

Component	Role
Discovery Galaxy	Renders the live, per-agent visualization of analysis progress and detailed findings.
PDF Report	Compiles the completed analysis into a downloadable, shareable document.

Table 1. System Architecture Components and Responsibilities

3. End-to-End Workflow

The flowchart below traces a single idea submission from the moment a user provides it through to the generation of the final PDF report, illustrating the complete, automated lifecycle of an analysis request.

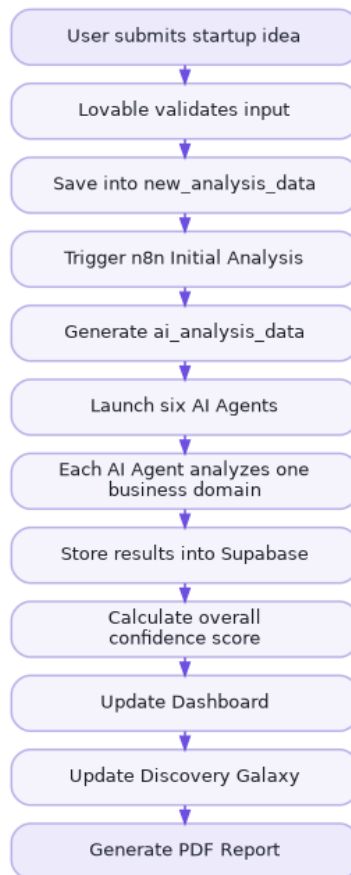


Figure 2. End-to-End Analysis Workflow

User submits startup idea. The user enters a startup concept, either as free text or a website URL, through the New Analysis screen in Lovable.

Lovable validates input. The front end checks that the submission is well-formed (for example, verifying URL structure) before it is sent onward, preventing malformed requests from reaching the AI pipeline.

Save into new_analysis_data. The validated submission is written to Supabase as a new record, establishing the permanent reference point for the rest of the workflow.

Trigger n8n Initial Analysis. A webhook call invokes the Initial Analysis workflow in n8n, passing the raw submission as its payload.

Generate ai_analysis_data. The Initial Analysis workflow calls the AI model to produce a refined title, description, industry classification, and matching score, then persists this as the master record.

Launch six AI Agents. Once the master record exists, the six specialist agent workflows are invoked, each operating independently.

Each AI Agent analyzes one business domain. Market Research, Competitor Analysis, Gap Analysis, Feasibility Analysis, Risk & Challenge, and Recommendation Engine each apply a domain-specific prompt to the same underlying idea.

Store results into Supabase. Each agent writes its structured findings to its own dedicated result table, referencing the original submission.

Calculate overall confidence score. Each agent independently derives a confidence score for its own findings, based on rules specific to that domain (see Section 8).

Update Dashboard. The Dashboard reads the latest status and scores across all tables to present a consolidated summary.

Update Discovery Galaxy. The Discovery Galaxy reflects live per-agent completion status as each workflow finishes.

Generate PDF Report. Once all agents have completed, the consolidated analysis can be compiled into a downloadable PDF report.

4. Initial Analysis Workflow (SS1)

Figure 3 shows the Initial Analysis workflow implemented in n8n. This workflow is the first automated process triggered after a user submits an idea: it receives the request from Lovable, validates it, calls the AI model to classify and describe the startup, and stores the resulting master record in ai_analysis_data.

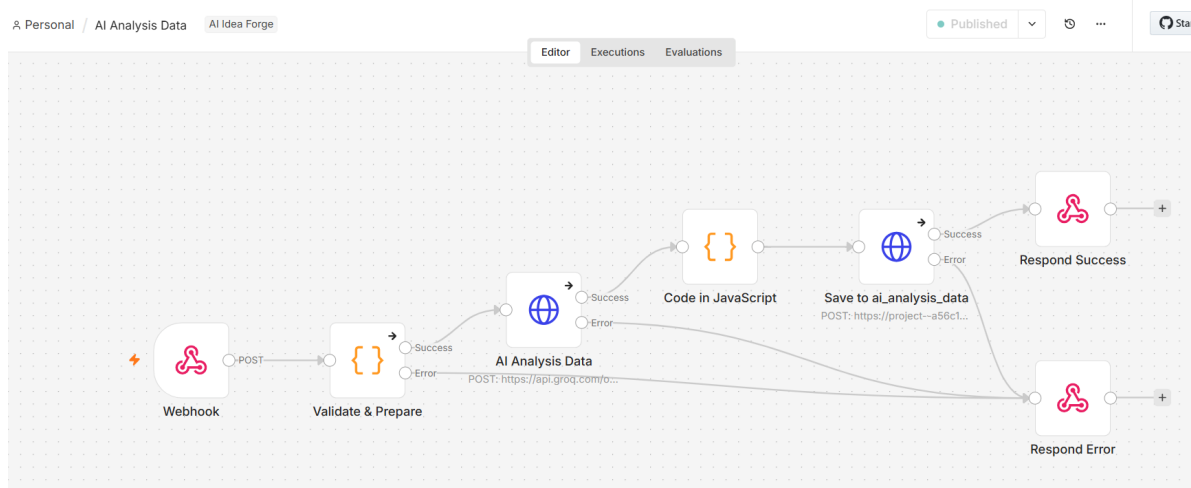


Figure 3. Initial AI Analysis Workflow (n8n)

Node	Purpose
Webhook	Receives the analysis request from Lovable as an HTTP POST call, triggering the workflow.

Node	Purpose
Validate & Prepare	Validates the incoming request payload and prepares it for the AI call, ensuring required fields are present.
AI Analysis Data	Calls the AI model (Llama 3.1 8B Instant via Groq) with a structured prompt to classify and describe the startup idea.
Code in JavaScript	Converts the AI model's raw text response into a structured JSON object matching the expected schema.
Save to ai_analysis_data	Persists the structured master analysis record to the ai_analysis_data table in Supabase via a REST call.
Respond Success	Returns a success response to Lovable once the record has been saved.
Respond Error	Returns a structured error response to Lovable if validation, the AI call, or the database save fails.

Table 2. Initial Analysis Workflow — Node Responsibilities

Each node in this workflow has a single, well-defined responsibility. The Webhook node acts as the workflow's entry point and is the only node directly reachable from Lovable. Validate & Prepare acts as a guard clause, ensuring the AI Analysis Data node is never called with an incomplete payload. The Code in JavaScript node exists because LLM responses are not guaranteed to arrive as clean JSON; this node normalizes the response before it reaches the database layer. Finally, the workflow branches into two distinct response paths — Respond Success and Respond Error — so that Lovable always receives a predictable, structured outcome regardless of where in the pipeline a failure occurs.

5. AI Agent Workflow (SS2)

Figure 4 shows the Market Research Agent workflow, which is representative of the workflow template used by all six specialist AI agents. Rather than building six separate, bespoke workflows, the platform uses a single reusable workflow structure that is cloned once per agent.

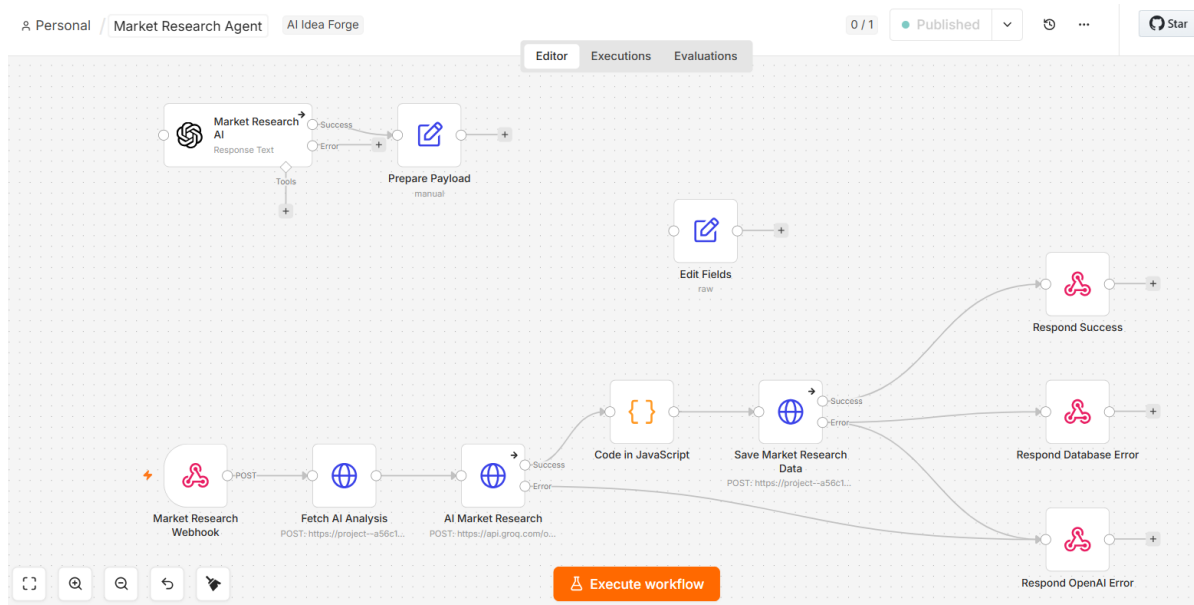


Figure 4. Reusable AI Agent Workflow (n8n) — Market Research Agent instance

Every AI agent workflow follows the identical node sequence — a webhook entry point, retrieval of the master analysis record, an AI call, JSON structuring, a database save, and one of three distinct response paths (success, database error, or AI/model error). Only four things differ between the six deployed instances of this template:

- **Workflow name** — for example, “Market Research Agent” versus “Risk & Challenge Agent.”
- **Prompt** — the system and user prompt sent to the AI model, which defines the domain the agent analyzes (see Section 8 and Appendix A).
- **Output JSON schema** — the specific fields the agent is required to return (see Appendix B).
- **Destination database table** — the Supabase table the structured result is saved to (for example, `market_research` versus `risk_challenge`).

This design is reusable and scalable because the orchestration logic — webhook handling, AI invocation, response parsing, database persistence, and error handling — is written once and proven correct once. Adding a new AI agent to the platform in the future does not require new orchestration logic to be designed or tested; it requires only cloning the existing template and substituting a new prompt, schema, and destination table. This also means that a bug fix or improvement to the shared workflow structure (for example, improved error messaging) can be propagated to every agent by updating the template, rather than requiring six separate code changes.

6. Generic AI Agent Workflow

The flowchart below abstracts the reusable AI agent workflow shown in Figure 4 into its generic node sequence, independent of any specific agent.

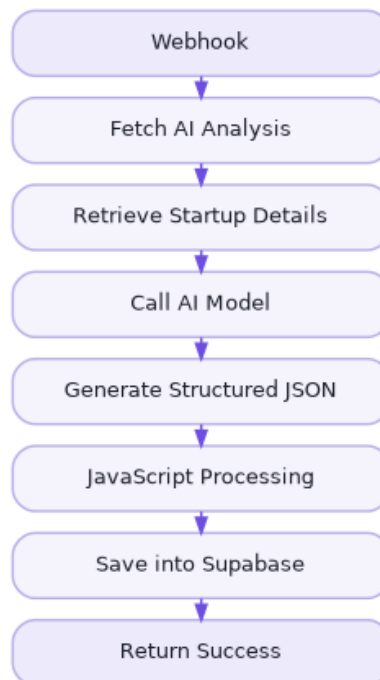


Figure 5. Generic AI Agent Workflow Template

Webhook. Serves as the entry point for the agent workflow, triggered once the master analysis record exists.

Fetch AI Analysis. Retrieves the ai_analysis_data master record for the idea being analyzed, using its identifier.

Retrieve Startup Details. Extracts the generated title, description, and industry classification from the master record for use in the agent's prompt.

Call AI Model. Sends the agent's domain-specific system and user prompt, populated with the startup's details, to the AI model.

Generate Structured JSON. The AI model returns its analysis as a JSON object matching the agent's required schema.

JavaScript Processing. Normalizes and validates the AI model's JSON response before it is persisted.

Save into Supabase. Writes the structured result to the agent's dedicated result table.

Return Success. Confirms completion of the agent's analysis back to the calling process.

7. AI Agent Responsibilities

AI Agent	Responsibility
Market Research	Evaluates market demand, industry trends, customer segments, and market size.
Competitor Analysis	Identifies competing products and analyzes market positioning.
Gap Analysis	Identifies market gaps, unmet customer needs, and innovation opportunities.
Feasibility Analysis	Evaluates business feasibility, technical complexity, budget, and timeline.
Risk & Challenge	Identifies business, technical, operational, financial, and regulatory risks.
Recommendation Engine	Generates strategic recommendations and actionable next steps.

Table 3. AI Agent Responsibilities by Business Domain

Each AI agent specializes in exactly one business domain rather than attempting a single, monolithic analysis. This specialization improves maintainability, since a prompt or schema change to one domain cannot unintentionally affect another; improves scalability, since new domains can be added as new agent instances without touching existing ones; and improves overall analysis quality, since a narrowly-scoped prompt produces more focused and consistent output than a single prompt attempting to cover six domains at once.

8. Individual AI Agent Documentation

This section documents each of the six specialist AI agents individually. For each agent, the workflow, data contract, and confidence scoring behaviour are described at the architectural level; the exact prompt text is not reproduced here but is provided in full in Appendix A.

8.1 Market Research

Purpose	Independently evaluates the market context surrounding the submitted startup idea.
Business Objective	Give the user an evidence-based view of market size, target customers, trends, and entry barriers before they invest further effort in the idea.
Workflow	Executed via the reusable AI Agent Workflow (Section 6 / Figure 5): Webhook → Fetch AI Analysis → Retrieve Startup Details → Call AI Model → Generate Structured JSON → JavaScript Processing → Save into Supabase (market_research) → Return Success.
Input	The AI-generated description from ai_analysis_data.
Output	A structured market research record covering problem definition, market sizing (TAM/SAM/SOM), target customers, trends, pain points, barriers to entry, growth rate, market maturity, and regulatory factors.
Prompt Objective	Constrains the AI model to act strictly as a market research consultant, explicitly excluding competitor analysis, recommendations, SWOT, feasibility, or risk content — keeping the agent's output scoped to market context only.
JSON Output Structure	<code>problem • market_size { tam, sam, som } • target_customers[] • market_trends[] • customer_pain_points[] • barriers_to_entry[] • market_growth_rate • market_maturity • regulatory_factors[] • sources_analyzed • confidence_score</code>
Database Table	market_research
Confidence Score Generation	Scored 50–100 based solely on market strength: 90–100 for a large, growing market with strong demand and manageable barriers; 76–89 for a good market with positive growth; 60–75 for a moderate market with uncertain demand; 50–59 for a small, saturated, or declining market.
Error Handling	The reusable template's Respond Database Error and Respond OpenAI Error paths return a structured error response if the Supabase save or the AI call fails, without corrupting the analysis record.

8.2 Competitor Analysis

Purpose	Independently evaluates the competitive landscape surrounding the submitted startup idea.
Business Objective	Help the user understand who they would be competing against and how the idea could be positioned relative to existing players.
Workflow	Executed via the reusable AI Agent Workflow (Section 6 / Figure 5): Webhook → Fetch AI Analysis → Retrieve Startup Details → Call AI Model → Generate Structured JSON → JavaScript Processing → Save into Supabase (competitor_analysis) → Return Success.
Input	The AI-generated title, description, and industry from ai_analysis_data.

Output	A structured competitor analysis record covering direct and indirect competitors, their common strengths and weaknesses, market positioning, and a competitive landscape summary.
Prompt Objective	Constrains the AI model to act strictly as a competitor analysis consultant, explicitly excluding market research, SWOT, recommendations, feasibility, or risk content.
JSON Output Structure	direct_competitors[] • indirect_competitors[] • competitor_strengths[] • competitor_weaknesses[] • market_positioning • competitive_landscape_summary • analysis_status • confidence_score • updated_at
Database Table	competitor_analysis
Confidence Score Generation	Scored 50–100 based on competitive saturation: 95–100 for very few competitors and a clear gap; 85–94 for moderate competition with good differentiation; 70–84 for a competitive market with identifiable gaps; 60–69 for a highly competitive market with limited differentiation; 50–59 for an extremely saturated market.
Error Handling	The reusable template's Respond Database Error and Respond OpenAI Error paths return a structured error response if the Supabase save or the AI call fails.

8.3 Gap Analysis

Purpose	Independently identifies unmet needs and innovation opportunities relevant to the submitted startup idea.
Business Objective	Surface concrete opportunities for the idea to differentiate itself, beyond what is captured by market sizing or competitor mapping alone.
Workflow	Executed via the reusable AI Agent Workflow (Section 6 / Figure 5): Webhook → Fetch AI Analysis → Retrieve Startup Details → Call AI Model → Generate Structured JSON → JavaScript Processing → Save into Supabase (gap_analysis) → Return Success.
Input	The AI-generated title, description, and industry from ai_analysis_data.
Output	A structured gap analysis record covering identified market gaps, unmet customer needs, innovation gaps, differentiation opportunities, and a gap summary.
Prompt Objective	Constrains the AI model to act strictly as a product innovation consultant, explicitly excluding market research, competitors, SWOT, feasibility, or risk content.
JSON Output Structure	identified_market_gaps[] • customer_unmet_needs[] • innovation_gaps[] • differentiation_opportunities[] • gap_summary • analysis_status • confidence_score
Database Table	gap_analysis
Confidence Score Generation	Scored 50–100 based on the strength of identified gaps: 90–100 for significant unmet needs and compelling differentiation; 76–89 for good gaps with reasonable differentiation; 60–75 for moderate, limited differentiation; 50–59 for few meaningful gaps.
Error Handling	The reusable template's Respond Database Error and Respond OpenAI Error paths return a structured error response if the Supabase save or the AI call fails.

8.4 Feasibility Analysis

Purpose	Independently evaluates how realistic the submitted startup idea is to build and commercialize.
Business Objective	Give the user a grounded view of technical and business feasibility before significant resources are committed.
Workflow	Executed via the reusable AI Agent Workflow (Section 6 / Figure 5): Webhook → Fetch AI Analysis → Retrieve Startup Details → Call AI Model → Generate Structured JSON → JavaScript Processing → Save into Supabase (feasibility_analysis) → Return Success.
Input	The AI-generated title, description, and industry from ai_analysis_data.
Output	A structured feasibility record covering business and technical feasibility scores, summaries, required resources, recommended technology stack, development complexity, time to market, and budget range.
Prompt Objective	Constrains the AI model to act strictly as a startup feasibility expert, explicitly excluding market research, competitors, SWOT, recommendations, or risk content.
JSON Output Structure	business_feasibility_score • technical_feasibility_score • overall_feasibility_score • business_feasibility_summary • technical_feasibility_summary • required_resources[] • technology_stack[] • estimated_development_complexity • estimated_time_to_market • estimated_budget_range • analysis_status • confidence_score
Database Table	feasibility_analysis
Confidence Score Generation	Scored 50–100 based on development complexity: 90–100 for Low complexity with strong viability; 76–89 for Medium complexity with good feasibility; 60–75 for High complexity with notable challenges; 50–59 for Very High complexity with major uncertainty.
Error Handling	The reusable template's Respond Database Error and Respond OpenAI Error paths return a structured error response if the Supabase save or the AI call fails.

8.5 Risk & Challenge

Purpose	Independently identifies the risks and operational challenges associated with the submitted startup idea.
Business Objective	Give the user a candid view of what could go wrong across market, technical, business, financial, legal, and operational dimensions.
Workflow	Executed via the reusable AI Agent Workflow (Section 6 / Figure 5): Webhook → Fetch AI Analysis → Retrieve Startup Details → Call AI Model → Generate Structured JSON → JavaScript Processing → Save into Supabase (risk_challenge) → Return Success.
Input	The AI-generated title, description, and industry from ai_analysis_data.
Output	A structured risk record covering market, technical, business, financial, and legal/regulatory risks, operational challenges, mitigation strategies, an overall risk level, and a risk summary.
Prompt Objective	Constrains the AI model to act strictly as a startup risk assessment expert, explicitly excluding market research, competitors, SWOT, recommendations, feasibility, or gap content.

JSON Output Structure	market_risks[] • technical_risks[] • business_risks[] • financial_risks[] • legal_and_regulatory_risks[] • operational_challenges[] • risk_mitigation_strategies[] • overall_risk_level • risk_summary • analysis_status • confidence_score
Database Table	risk_challenge
Confidence Score Generation	Scored 50–100 based on overall risk level: 76–100 for Low risk; 65–85 for Medium risk; 50–75 for High or Very High risk.
Error Handling	The reusable template's Respond Database Error and Respond OpenAI Error paths return a structured error response if the Supabase save or the AI call fails.

8.6 Recommendation Engine

Purpose	Synthesizes strategic recommendations for the submitted startup idea.
Business Objective	Translate the idea into a concrete, prioritized set of next steps and strategic direction the user can act on.
Workflow	Executed via the reusable AI Agent Workflow (Section 6 / Figure 5): Webhook → Fetch AI Analysis → Retrieve Startup Details → Call AI Model → Generate Structured JSON → JavaScript Processing → Save into Supabase (recommendation_engine) → Return Success.
Input	The AI-generated title, description, and industry from ai_analysis_data.
Output	A structured recommendation record covering an overall opportunity score, priority recommendations, next steps, product improvements, business strategy, go-to-market strategy, and a recommendation summary.
Prompt Objective	Constrains the AI model to act strictly as a startup strategy advisor, explicitly excluding market research, competitors, SWOT, feasibility, or risk content.
JSON Output Structure	overall_opportunity_score • priority_recommendations[] • recommended_next_steps[] • product_improvements[] • business_strategy[] • go_to_market_strategy[] • recommendation_summary • analysis_status • confidence_score
Database Table	recommendation_engine
Confidence Score Generation	Scored 50–100 based on the strength of the opportunity: 90–100 for a very strong opportunity with high-impact recommendations; 76–89 for a good opportunity; 60–75 for a moderate opportunity with uncertainties; 50–59 for weak strategic potential.
Error Handling	The reusable template's Respond Database Error and Respond OpenAI Error paths return a structured error response if the Supabase save or the AI call fails.

9. Lovable ↔ n8n Integration

The sequence diagram below illustrates the runtime interaction between the user, the Lovable front end, the n8n orchestration layer, the AI model, Supabase, and the platform's downstream outputs, for a single analysis request.

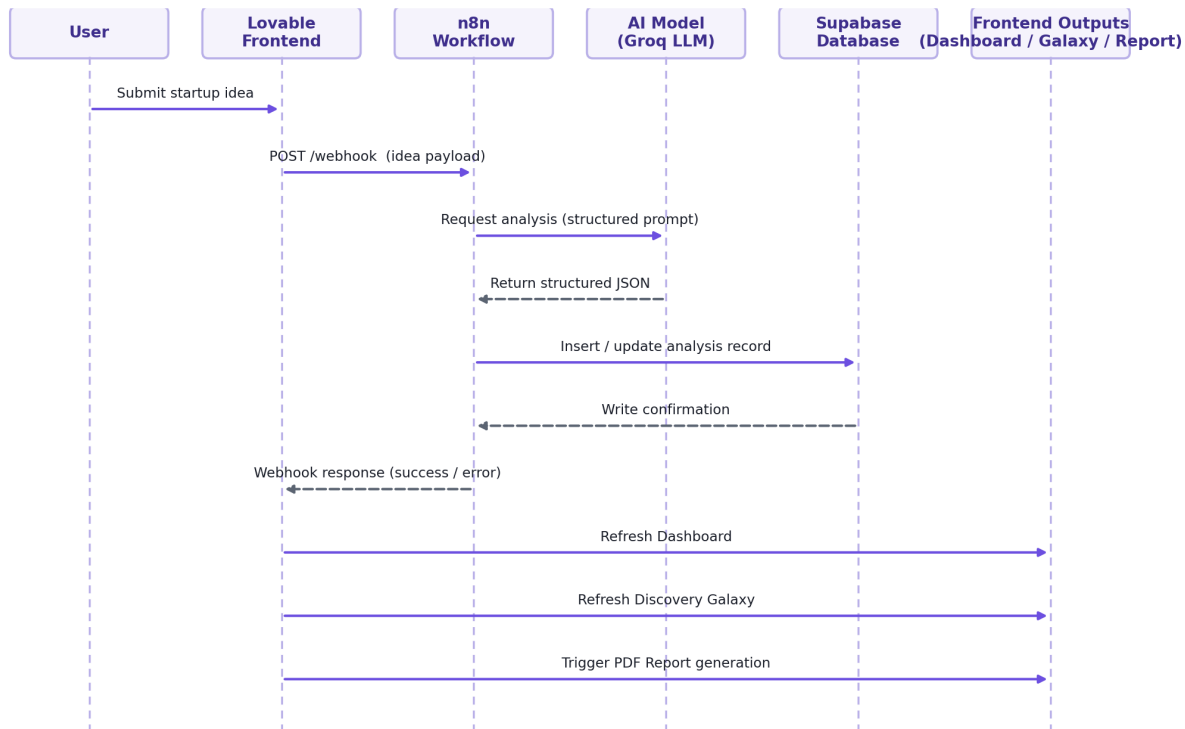


Figure 6. Sequence Diagram – Lovable ↔ n8n Integration

HTTP communication and API requests. Lovable and n8n communicate exclusively over HTTP. Lovable never calls the AI model or Supabase directly for the analysis pipeline; every request is routed through an n8n webhook, which keeps the orchestration logic centralized in one layer.

Webhooks. Each n8n workflow exposes a webhook URL as its trigger. Lovable invokes these webhooks with a POST request whenever a new idea is submitted or a new agent needs to run, and the corresponding workflow begins executing immediately upon receipt.

JSON payload. Every request and response in this integration is a JSON document. Lovable sends the raw idea (or, for agent workflows, a reference to the existing analysis) as the request body, and the AI model's own output is likewise constrained to JSON so that it can be parsed deterministically downstream.

Response handling. Each n8n workflow ends in one of a small number of well-defined response nodes (see Sections 4 and 5), so Lovable always receives a predictable success or error payload rather than having to infer the outcome. Lovable uses this response to update its own local state and refresh the Dashboard and Discovery Galaxy views.

10. n8n ↔ Supabase Integration

The workflow below details how each n8n AI workflow interacts with Supabase to read the data it needs and persist its results.

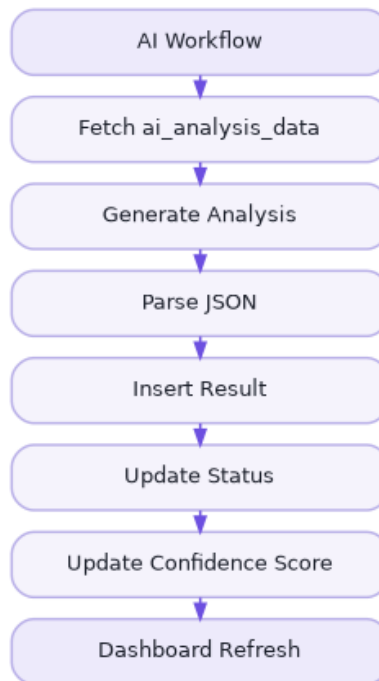


Figure 7. n8n ↔ Supabase Integration Workflow

Fetch operations. Before an agent can analyze an idea, it must first fetch the `ai_analysis_data` master record via a Supabase REST call, retrieving the AI-generated title, description, and industry that will populate its prompt.

Insert operations. Once the AI model returns its structured JSON, the workflow inserts a new row into the agent's dedicated result table (for example, `market_research` or `risk_challenge`), using the originating analysis identifier as a foreign key.

Update operations. Alongside the insert, the workflow updates status fields — both on the newly created result row and, where applicable, on the parent record — to reflect that the agent has completed.

Confidence synchronization. Each agent's confidence score is written directly to its own result row at the same time as its findings, so the Dashboard and Discovery Galaxy can display confidence per agent without any additional reconciliation step.

Status synchronization. Because each agent table carries its own `analysis_status` field, the front end can determine overall completion by checking the status of all six tables against a single originating analysis identifier, without requiring a separate orchestration table.

11. Workflow Automation

Concept	Description
Workflow Automation	n8n automates the entire AI analysis pipeline.
AI Orchestration	Coordinates six independent AI agents.
Sequential Processing	Executes dependent workflows in order.
Data Synchronization	Synchronizes Lovable, Supabase, and n8n.
Status Updates	Maintains Pending, Running, Completed, and Failed states.
Workflow Completion	Finalizes analysis after all six AI agents complete successfully.

Table 4. Core Workflow Automation Concepts

Together, these concepts describe a pipeline that is automated end-to-end but remains observable at every stage: each analysis request progresses through explicit states, and the platform can always report which of the six agents — if any — are still pending, rather than treating the analysis as a single opaque operation.

12. Error Handling

The flowchart below shows the error handling pattern used consistently across both the Initial Analysis workflow and every AI agent workflow.

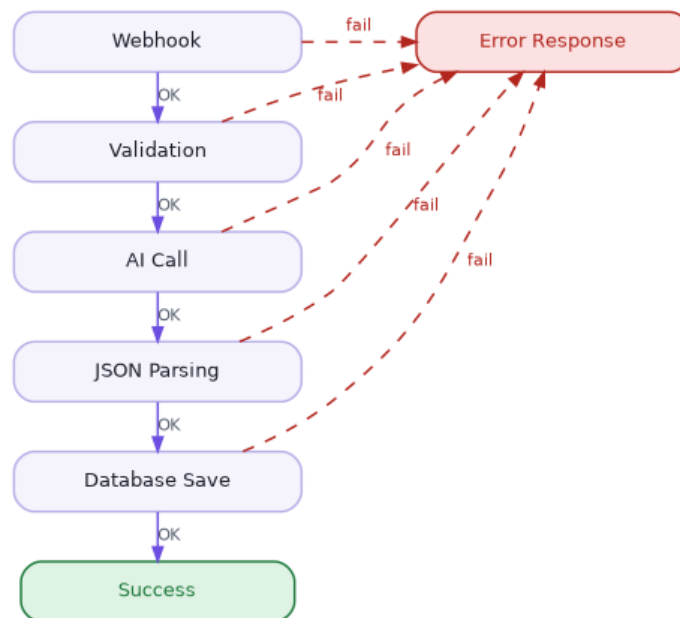


Figure 8. Error Handling Flowchart

Validation errors. If the incoming payload is missing required fields or is malformed, the workflow short-circuits at the Validation stage and returns an error response without attempting an AI call.

AI failures. If the call to the AI model fails or times out, the workflow's dedicated OpenAI/model error path returns a structured error response, as seen in the Respond OpenAI Error node in Figure 4.

JSON parsing failures. Because the AI model's output is free-form text constrained only by prompt instructions, the JavaScript processing stage may encounter a response that fails to parse as valid JSON; this is treated as a failure of the AI Call stage rather than being allowed to propagate an invalid record to the database.

Database failures. If the Supabase save fails — for example, due to a constraint violation or connectivity issue — the workflow's Respond Database Error path returns a structured error without leaving a partially-written record behind.

Retry mechanism and recovery strategy. The workflows themselves are not currently observed to perform automatic retries internally; instead, recovery is handled at the application level — Lovable's Resume Analysis capability allows a user to re-enter a pending or failed analysis and have the platform pick up from its last known state rather than restarting the entire pipeline. This keeps the n8n workflows simple and stateless, while placing recovery responsibility in the layer that already tracks per-agent status.

13. Design Principles

Principle	Description
Modular Architecture	Independent AI workflows
Separation of Responsibilities	Dedicated AI agent for each business domain
Workflow Reusability	Common workflow reused by six agents
Loose Coupling	Components communicate through APIs
Scalability	Easy to add future AI agents
Maintainability	Independent workflow updates
Fault Tolerance	Failure isolation
Extensibility	Supports additional AI workflows

Table 5. Architectural Design Principles

14. Conclusion

The AI Idea Forge platform's integration architecture cleanly separates presentation, orchestration, intelligence, and persistence. Lovable provides the user interface through which ideas are submitted and results are reviewed. n8n orchestrates every AI workflow in the pipeline, from the initial classification of an idea through to six parallel, domain-specific analyses, using a single reusable workflow template to keep the system maintainable as it grows. The AI model performs the actual domain-specific reasoning within tightly-scoped, structured prompts. Supabase stores every stage of the analysis in normalized, purpose-built tables. Finally, the Dashboard and Discovery Galaxy consume this structured data to give the user both an at-a-glance summary and a detailed, explorable view of their idea's analysis. Together, these layers form an automation architecture that is modular, observable, and extensible to future AI agents.

Appendix A – AI Prompt Engineering

Each AI workflow in the platform uses structured prompt engineering to ensure consistent AI responses, strictly structured JSON output, domain-specific analysis scoped to a single business concern, reliable downstream processing, and easy parsing and storage. Every agent's system prompt establishes its professional persona and output constraints, while its user prompt supplies the JSON schema, field-level rules, confidence-scoring guidance, and the startup idea itself. The complete prompt text for each of the six AI agents, exactly as implemented in n8n, is reproduced below.

A.1 Market Research Prompt

System Prompt

You are a professional Market Research Consultant. Always return ONLY valid JSON. Never use markdown, code fences, explanations, comments, or additional text. Follow the JSON schema exactly. Generate ONLY market research. Never include competitors, recommendations, SWOT analysis, opportunities, business model, feasibility analysis, implementation planning, risk analysis, or gap analysis.

User Prompt

Analyze the following startup idea and generate ONLY market research.

Return ONLY this JSON structure:

```
{
  "problem": "",
  "market_size": {
    "tam": "",
    "sam": "",
    "som": ""
  },
  "target_customers": [],
  "market_trends": [],
  "customer_pain_points": [],
  "barriers_to_entry": [],
  "market_growth_rate": "",
  "market_maturity": "",
  "regulatory_factors": [],
  "sources_analyzed": 0,
  "confidence_score": 0
}
```

Rules:

- problem MUST describe the primary business problem the startup solves.
- market_size MUST include realistic TAM, SAM, and SOM estimates with values and units.
- target_customers MUST be an array of customer segments.
- market_trends MUST contain 4-6 current market trends.
- customer_pain_points MUST contain 4-6 customer pain points.
- barriers_to_entry MUST contain 4-6 market entry barriers.
- market_growth_rate MUST provide an estimated CAGR or annual growth percentage.
- market_maturity MUST be one of: Emerging, Growing, Mature, or Saturated.
- regulatory_factors MUST list applicable regulations, compliance requirements, or legal considerations.
- sources_analyzed MUST be an integer representing the approximate number of credible market sources considered.

Confidence Score Rules:

- confidence_score MUST be an integer between 50 and 100.
- Assign the confidence score based ONLY on the market research.
- If the startup targets a large and growing market, has strong customer demand, favorable trends, manageable barriers to entry, and reliable market data, assign a confidence score between 90 and 100.
- If the startup targets a good market with positive growth, clear customer demand, and moderate barriers to entry, assign a confidence score between 76 and 89.
- If the startup targets a moderate market with average growth, uncertain demand, or notable barriers to entry, assign a confidence score between 60 and 75.
- If the startup targets a small, saturated, or declining market with weak demand, significant barriers, or limited supporting data, assign a confidence score between 50 and 59.
- Choose the score that best matches the market research. Do NOT use a default value.

IMPORTANT:

- Generate ONLY market research.

- DO NOT include competitors.
- DO NOT include SWOT analysis.
- DO NOT include recommendations.
- DO NOT include opportunities.
- DO NOT include business model.
- DO NOT include feasibility analysis.
- DO NOT include implementation planning.
- DO NOT include risk analysis.
- DO NOT include gap analysis.
- Return ONLY the JSON object without any additional text.

Startup Idea:

```
{{ $json.generated_description }}
```

A.2 Competitor Analysis Prompt

System Prompt

You are a professional Competitor Analysis Consultant. Always return ONLY valid JSON. Never use markdown, code fences, explanations, comments, or additional text. Follow the JSON schema exactly. Generate ONLY competitor analysis. Never include market research, SWOT analysis, recommendations, business model, feasibility analysis, implementation planning, risk analysis, or gap analysis.

User Prompt

Analyze the following startup idea and generate ONLY competitor analysis.

Return ONLY this JSON structure:

```
{
  "direct_competitors": [],
  "indirect_competitors": [],
  "competitor_strengths": [],
  "competitor_weaknesses": [],
  "market_positioning": "",
  "competitive_landscape_summary": "",
  "analysis_status": "Completed",
  "confidence_score": 0,
  "updated_at": ""
}
```

Rules:

- direct_competitors MUST contain 4-8 well-known direct competitors relevant to the startup idea.
- indirect_competitors MUST contain 4-8 indirect competitors or alternative solutions.
- competitor_strengths MUST contain 5-8 common strengths observed across competitors.
- competitor_weaknesses MUST contain 5-8 common weaknesses or limitations observed across competitors.
- market_positioning MUST describe how the startup could position itself within the competitive landscape in 2-4 sentences.
- competitive_landscape_summary MUST summarize the overall competitive landscape in 2-4 sentences.
- analysis_status MUST always be "Completed".

Confidence Score Rules:

- confidence_score MUST be an integer between 50 and 100.
- Assign the confidence score based ONLY on the competitor analysis.

Scoring Guide:

- 95-100: Very few competitors, clear market gap, strong differentiation, low market saturation, and strong competitive advantage.
- 85-94: Moderate competition with good differentiation, clear positioning, and realistic opportunity to capture market share.
- 70-84: Competitive market with several established players but identifiable gaps and reasonable opportunities for success.
- 60-69: Highly competitive market with limited differentiation. Success depends heavily on execution and marketing.
- 50-59: Extremely saturated market dominated by strong competitors with little or no clear competitive advantage.
- Do NOT select a default value such as 90 or 92. Choose the confidence score that best matches the analysis.
- updated_at MUST be the current timestamp in ISO 8601 format.

IMPORTANT:

- Generate ONLY competitor analysis.
- DO NOT include market research.
- DO NOT include SWOT analysis.
- DO NOT include recommendations.
- DO NOT include business model.
- DO NOT include feasibility analysis.
- DO NOT include implementation planning.
- DO NOT include risk analysis.
- DO NOT include gap analysis.
- Return ONLY the JSON object without any additional text.

Startup Idea:

Title:
{{ \$json.generated_title }}

Description:
{{ \$json.generated_description }}

Industry:
{{ \$json.industry }}

A.3 Gap Analysis Prompt

System Prompt

You are a professional Product Innovation Consultant. Always return ONLY valid JSON. Never use markdown, code fences, explanations, comments, or additional text. Follow the JSON schema exactly. Generate ONLY gap analysis. Never include market research, competitors, recommendations, SWOT analysis, business model, feasibility analysis, implementation planning, or risk analysis.

User Prompt

Analyze the following startup idea and generate ONLY gap analysis.

Return ONLY this JSON structure:

```
{
  "identified_market_gaps": [],
  "customer_unmet_needs": [],
  "innovation_gaps": [],
  "differentiation_opportunities": [],
  "gap_summary": "",
  "analysis_status": "Completed",
  "confidence_score": 0
}
```

Rules:

- identified_market_gaps MUST contain 4-6 specific gaps in the current market.
- customer_unmet_needs MUST contain 4-6 unmet customer needs that existing solutions fail to address.
- innovation_gaps MUST contain 4-6 opportunities where innovation can improve existing solutions.
- differentiation_opportunities MUST contain 4-6 unique ways the startup could differentiate itself from existing market offerings without naming competitors.
- gap_summary MUST provide a concise summary of the overall market gap opportunity in 2-4 sentences.
- analysis_status MUST always be "Completed".

Confidence Score Rules:

- confidence_score MUST be an integer between 50 and 100.
- Assign the confidence score based ONLY on the gap analysis.
- If the startup addresses significant unmet customer needs, identifies strong market gaps, offers clear innovation opportunities, and has compelling differentiation, assign a confidence score between 90 and 100.
- If the startup identifies good market gaps with reasonable innovation and differentiation opportunities, assign a confidence score between 76 and 89.
- If the startup identifies only moderate gaps with limited differentiation or innovation potential, assign a confidence score between 60 and 75.
- If the startup identifies few meaningful market gaps, weak innovation opportunities, or minimal differentiation potential, assign a confidence score between 50 and 59.
- Choose the score that best matches the gap analysis. Do NOT use a default value.

IMPORTANT:

- Generate ONLY gap analysis.
- DO NOT include market research.
- DO NOT include competitors.
- DO NOT include SWOT analysis.
- DO NOT include recommendations.
- DO NOT include business model.
- DO NOT include feasibility analysis.
- DO NOT include implementation planning.
- DO NOT include risk analysis.
- Return ONLY the JSON object without any additional text.

Startup Idea:

Title:
{{ \$json.generated_title }}

Description:
{{ \$json.generated_description }}

Industry:
{{ \$json.industry }}

A.4 Feasibility Analysis Prompt

System Prompt

You are a professional Startup Feasibility Expert. Always return ONLY valid JSON. Never use markdown, code fences, explanations, comments, or additional text. Follow the JSON schema exactly. Generate ONLY feasibility analysis. Never include market research, competitors, recommendations, SWOT analysis, business model, implementation planning, risk analysis, or gap analysis.

User Prompt

Analyze the following startup idea and generate ONLY feasibility analysis.

Return ONLY this JSON structure:

```
{
  "business_feasibility_score": 0,
  "technical_feasibility_score": 0,
  "overall_feasibility_score": 0,
  "business_feasibility_summary": "",
  "technical_feasibility_summary": "",
  "required_resources": [],
  "technology_stack": [],
  "estimated_development_complexity": "",
  "estimated_time_to_market": "",
  "estimated_budget_range": "",
  "analysis_status": "Completed",
  "confidence_score": 0
}
```

Rules:

- business_feasibility_score MUST be an integer between 0 and 100.
- technical_feasibility_score MUST be an integer between 0 and 100.
- overall_feasibility_score MUST be an integer between 0 and 100.
- business_feasibility_summary MUST summarize commercial viability in 2-4 sentences.
- technical_feasibility_summary MUST summarize technical viability in 2-4 sentences.
- required_resources MUST contain 5-8 key resources required to build and launch the startup.
- technology_stack MUST contain 5-10 recommended technologies, platforms, frameworks, or tools required for implementation.
- estimated_development_complexity MUST be one of: Low, Medium, High, or Very High.
- estimated_time_to_market MUST provide a realistic estimated development timeline.
- estimated_budget_range MUST provide a realistic estimated budget range in USD.
- analysis_status MUST always be "Completed".

Confidence Score Rules:

- confidence_score MUST be an integer between 50 and 100.
- Assign the confidence score based ONLY on the feasibility analysis.

- If the startup has Low development complexity, high business viability, realistic budget, achievable timeline, and a strong technical foundation, assign a confidence score between 90 and 100.
- If the startup has Medium development complexity with good business and technical feasibility, assign a confidence score between 76 and 89.
- If the startup has High development complexity, significant technical challenges, higher budget, or longer time to market, assign a confidence score between 60 and 75.
- If the startup has Very High development complexity, major technical uncertainty, unrealistic budget or timeline, or poor feasibility, assign a confidence score between 50 and 59.
- Choose the score that best matches the feasibility analysis. Do NOT use a default value.

IMPORTANT:

- Generate ONLY feasibility analysis.
- DO NOT include market research.
- DO NOT include competitors.
- DO NOT include SWOT analysis.
- DO NOT include recommendations.
- DO NOT include business model.
- DO NOT include implementation planning.
- DO NOT include risk analysis.
- DO NOT include gap analysis.
- Return ONLY the JSON object without any additional text.

Startup Idea:

Title:

```
{{ $json.generated_title }}
```

Description:

```
{{ $json.generated_description }}
```

Industry:

```
{{ $json.industry }}
```

A.5 Risk & Challenge Prompt

System Prompt

You are a professional Startup Risk Assessment Expert. Always return ONLY valid JSON. Never use markdown, code fences, explanations, comments, or additional text. Follow the JSON schema exactly. Generate ONLY startup risk analysis. Never include market research, competitors, recommendations, SWOT analysis, business model, feasibility analysis, implementation planning, or gap analysis.

User Prompt

Analyze the following startup idea and generate ONLY startup risk analysis.

Return ONLY this JSON structure:

```
{
  "market_risks": [],
  "technical_risks": [],
  "business_risks": [],
  "financial_risks": [],
  "legal_and_regulatory_risks": [],
  "operational_challenges": [],
  "risk_mitigation_strategies": [],
  "overall_risk_level": "",
  "risk_summary": "",
  "analysis_status": "Completed",
  "confidence_score": 0
}
```

Rules:

- market_risks MUST contain 4-6 market-related risks.
- technical_risks MUST contain 4-6 technical implementation risks.
- business_risks MUST contain 4-6 business strategy or execution risks.
- financial_risks MUST contain 4-6 financial risks.
- legal_and_regulatory_risks MUST contain 3-5 applicable legal, compliance, or regulatory risks.
- operational_challenges MUST contain 4-6 operational or scaling challenges.
- risk_mitigation_strategies MUST contain 5-8 practical mitigation strategies corresponding to the identified

risks.

- overall_risk_level MUST be one of: Low, Medium, High, or Very High.
- risk_summary MUST summarize the startup's overall risk profile in 2-4 sentences.
- analysis_status MUST always be "Completed".

Confidence Score Rules:

- confidence_score MUST be an integer between 50 and 100.
- Assign the confidence score based ONLY on the overall risk assessment.
- If the startup has Low risk, assign a confidence score between 76 and 100.
- If the startup has Medium risk, assign a confidence score between 65 and 85.
- If the startup has High or Very High risk, assign a confidence score between 50 and 75.
- Choose the score that best matches the identified risks and mitigation strategies. Do NOT use a default value.

IMPORTANT:

- Generate ONLY startup risk analysis.
- DO NOT include market research.
- DO NOT include competitors.
- DO NOT include SWOT analysis.
- DO NOT include recommendations.
- DO NOT include business model.
- DO NOT include feasibility analysis.
- DO NOT include implementation planning.
- DO NOT include gap analysis.
- Return ONLY the JSON object without any additional text.

Startup Idea:

Title:

{{ \$json.generated_title }}

Description:

{{ \$json.generated_description }}

Industry:

{{ \$json.industry }}

A.6 Recommendation Engine Prompt

System Prompt

You are a professional Startup Strategy Advisor. Always return ONLY valid JSON. Never use markdown, code fences, explanations, comments, or additional text. Follow the JSON schema exactly. Generate ONLY strategic recommendations. Never include market research, competitors, SWOT analysis, feasibility analysis, implementation planning, risk analysis, or gap analysis.

User Prompt

Analyze the following startup idea and generate ONLY strategic recommendations.

Return ONLY this JSON structure:

```
{
  "overall_opportunity_score": 0,
  "priority_recommendations": [],
  "recommended_next_steps": [],
  "product_improvements": [],
  "business_strategy": [],
  "go_to_market_strategy": [],
  "recommendation_summary": "",
  "analysis_status": "Completed",
  "confidence_score": 0
}
```

Rules:

- overall_opportunity_score MUST be an integer between 0 and 100.
- priority_recommendations MUST contain 5-8 high-priority strategic recommendations.
- recommended_next_steps MUST contain 5-8 actionable next steps in logical order.
- product_improvements MUST contain 4-6 suggested product enhancements.
- business_strategy MUST contain 4-6 strategic business initiatives.
- go_to_market_strategy MUST contain 4-6 go-to-market strategies.

- recommendation_summary MUST summarize the overall strategic direction in 2-4 sentences.
- analysis_status MUST always be "Completed".

Confidence Score Rules:

- confidence_score MUST be an integer between 50 and 100.
- Assign the confidence score based ONLY on the quality and strength of the strategic recommendations.
- If the startup has a very strong opportunity with clear, practical, and high-impact recommendations, assign a confidence score between 90 and 100.
- If the startup has a good opportunity with actionable recommendations and a realistic growth path, assign a confidence score between 76 and 89.
- If the startup has a moderate opportunity with several challenges or uncertainties, assign a confidence score between 60 and 75.
- If the startup has weak strategic potential, unclear direction, or significant obstacles, assign a confidence score between 50 and 59.
- Choose the score that best matches the analysis. Do NOT use a default value.

IMPORTANT:

- Generate ONLY strategic recommendations.
- DO NOT include market research.
- DO NOT include competitors.
- DO NOT include SWOT analysis.
- DO NOT include feasibility analysis.
- DO NOT include implementation planning.
- DO NOT include risk analysis.
- DO NOT include gap analysis.
- Return ONLY the JSON object without any additional text.

Startup Idea:

Title:

{{ \$json.generated_title }}

Description:

{{ \$json.generated_description }}

Industry:

{{ \$json.industry }}

Appendix B – JSON Output Schemas

The JSON structures below show the exact schema each AI agent is required to return, as defined in its user prompt. Field-level meaning is described in Section 8; this appendix presents the schemas themselves as compact reference blocks.

B.1 Market Research — saved to `market_research`

```
{
  "problem": "",
  "market_size": {
    "tam": "",
    "sam": "",
    "som": ""
  },
  "target_customers": [],
  "market_trends": [],
  "customer_pain_points": [],
  "barriers_to_entry": [],
  "market_growth_rate": "",
  "market_maturity": "",
  "regulatory_factors": [],
  "sources_analyzed": 0,
  "confidence_score": 0
}
```

B.2 Competitor Analysis — saved to `competitor_analysis`

```
{
  "direct_competitors": [],
  "indirect_competitors": [],
  "competitor_strengths": [],
  "competitor_weaknesses": [],
  "market_positioning": "",
  "competitive_landscape_summary": "",
  "analysis_status": "Completed",
  "confidence_score": 0,
  "updated_at": ""
}
```

B.3 Gap Analysis — saved to `gap_analysis`

```
{
  "identified_market_gaps": [],
  "customer_unmet_needs": [],
  "innovation_gaps": [],
  "differentiation_opportunities": [],
  "gap_summary": "",
  "analysis_status": "Completed",
  "confidence_score": 0
}
```

B.4 Feasibility Analysis — saved to `feasibility_analysis`

```
{
  "business_feasibility_score": 0,
  "technical_feasibility_score": 0,
  "overall_feasibility_score": 0,
  "business_feasibility_summary": "",
  "technical_feasibility_summary": "",
  "required_resources": [],
  "technology_stack": [],
  "estimated_development_complexity": "",
  "estimated_time_to_market": "",
  "estimated_budget_range": "",
  "analysis_status": "Completed",
  "confidence_score": 0
}
```

B.5 Risk & Challenge — saved to risk_challenge

```
{
  "market_risks": [],
  "technical_risks": [],
  "business_risks": [],
  "financial_risks": [],
  "legal_and_regulatory_risks": [],
  "operational_challenges": [],
  "risk_mitigation_strategies": [],
  "overall_risk_level": "",
  "risk_summary": "",
  "analysis_status": "Completed",
  "confidence_score": 0
}
```

B.6 Recommendation Engine — saved to recommendation_engine

```
{
  "overall_opportunity_score": 0,
  "priority_recommendations": [],
  "recommended_next_steps": [],
  "product_improvements": [],
  "business_strategy": [],
  "go_to_market_strategy": [],
  "recommendation_summary": "",
  "analysis_status": "Completed",
  "confidence_score": 0
}
```